Further applications of the normal distribution

1 The operating times of phone batteries are approximately normally distributed with a mean of 34 hours and a standard deviation of 4 hours. Answer the following questions using the empirical rule:

- **a** What percentage of batteries last between 33 and 38 hours? Round your answer to the nearest percent.
- **b** What percentage of batteries last between 28 hours and 41 hours?
- **c** Any battery that lasts less than 23 hours is deemed faulty. If a company manufactured $51\,000$ batteries, approximately how many batteries would they be able to sell? Round your answer to the nearest integer.

2 The height of sunflowers is approximately normally distributed, with a mean height of $1.6\ \text{m}$ and a standard deviation of $8\ \text{cm}$.

- **a** Approximately what percentage of sunflowers are between $1.55\ \text{m}$ and $1.60\ \text{m}$ tall? Round your answer to the nearest percent.
- **b** Approximately what percentage of sunflowers are between $1.69\ \text{m}$ and $1.81\ \text{m}$ tall? Round your answer to the nearest percent.
- **c** If there are 3000 sunflowers in the field, approximately how many are taller than $1.52\ \text{m}$? Round your answer to the nearest integer.

3 The length of the tail of a domestic cat is normally distributed with a mean of $25\ \text{cm}$ and a standard deviation of $2.2\ \text{cm}$. Use your graphics calculator to answer the following, rounding your answers to three decimal places:

- **a** What is the shortest length of a tail in the 70th percentile?
- **b** What is the shortest length of a tail in the top 15%?
- **c** What is the shortest length of a tail in the 0.45 quantile?
- **d** What is the probability that a cat has a tail length less than $23.5\ \text{cm}$?
- **e** Suppose that a cat has a tail length below the 80th percentile. What is the probability that their tail length is more than $23.5\ \text{cm}$?

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4 The weights of babies born in a hospital in Sydney are considered normally distributed with a mean of $3.3\ \text{kg}$ and a standard deviation of $0.50\ \text{kg}$. Use your graphics calculator to answer the following, rounding your answers to three decimal places:

a What is the weight of the smallest baby in the 65th percentile?

b What is the weight of the smallest baby in the top 20%?

c What is the weight of the smallest baby in the 0.55 quantile?

d What is the probability that a baby weighs less than $3.05\ \text{kg}$? Give your answer as a decimal.

e Suppose that a baby has a weight below the 70th percentile. What is the probability that their weight is more than $3.05\ \text{kg}$?

5 People in the Dinaric Alps are considered the tallest in the world. The heights of males are considered to be normally distributed with a mean height of $186.0\ \text{cm}$ and a standard deviation of $6\ \text{cm}$. Use your graphics calculator to answer the following, rounding your answers to three decimal places unless otherwise stated:

a What is the height of the shortest male in the top 25%?

b What is the height of the tallest male in the 0.95 quantile?

c What percentage of males are shorter than $176\ \text{cm}$? Round your answer to the nearest tenth of a percent.

d What is the probability that a male has a height less than $187\ \text{cm}$?

e Suppose that a male has a height below the 90th percentile. What is the probability that their height is more than $187\ \text{cm}$?

6 A machine is set for the production of cylinders of mean diameter $5.06\ \text{cm}$, with standard deviation $0.016\ \text{cm}$.

a Assuming a normal distribution, between what values, in centimetres, will 99.7% of the diameters lie?

b If cylinders with diameters less than $5.012\ \text{cm}$ or more than $5.108\ \text{cm}$ are discarded, what percentage of cylinders produced are discarded?

c If a cylinder, randomly selected from this production, has a diameter of $5.124\ \text{cm}$, what conclusion could be drawn?

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7 At Summer Heights High, 148 students are studying Physics. The results on the first semester examination saw a mean mark of 60% with a standard deviation of 11.4%, and the results were considered to be approximately normally distributed.

a Luke scored 63% on the first semester exam. What percentage of students scored more than Luke on the first semester exam? Round your answer to the nearest percentage.

b Students whose score was more than two standard deviations below the mean were advised not to continue with the study of Physics. At most, how many students were given this advice?

c The results of the end of year examination saw a mean mark of 67% with a standard deviation of 9.3%. If Luke did just as well in the end of year examination, relative to his peers, as he did in the first semester examination, then what score would Luke receive?

8 Victoria downloads each episode of her favourite TV show as it's released online. The length of each show is represented by the random variable T, which is approximately normally distributed with a mean length of 50 minutes and a standard deviation of 4 minutes.

a What percentage of her shows are less than 49 minutes in length?

b Victoria wants to put a show on her USB drive but only has room for an episode that is 48 minutes in length. What is the probability that she won't be able to fit the show on the drive?

c Of the next five shows that Victoria independently downloads, what is the probability that the first two are less than 49 minutes and the last three are more than 49 minutes?

d Of the next 5 shows that Victoria downloads, what is the probability that exactly two are less than 49 minutes?

e Fans of the show have complained that the show length is really inconsistent. Calculate the maximum value of the standard deviation such that the probability of a show being less than 45 minutes is no more than 0.2%.

9 The raw exam results, X, for an ATAR subject is approximately normally distributed with a mean of 56% and a standard deviation of 13%.

a If 2700 students sat the exam, what is the lowest score corresponding to the top 54 scoring students?

b If 14 results are chosen at random for this subject, what is the probability that at most 5 students scored less than 51%?

10 A packet of crisps is advertised as weighing 175 g, but the actual packets are normally distributed with a mean mass of 174.8 g and a standard deviation of 1.2 g.

a What is the probability that a randomly selected packet weighs more than the advertised weight?

b What is the largest weight out of the lightest 5% of packets?

c Out of 11 randomly selected packets, what is the probability that at least 5 packets weigh more than 174.6 g?

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